Network Traffic Analysis: DNS Service Outage

Investigating a “destination port unreachable” error using tcpdump

Type	Cybersecurity incident report (network traffic analysis)
Tools	tcpdump
Protocols	DNS, UDP, ICMP, HTTPS
Skills	Packet capture analysis, TCP/IP model, incident documentation, root-cause hypotheses
Source	Google Cybersecurity Professional Certificate, Course 3 (*Connect and Protect: Networks and Network Security*)

Situation

Customers of a client reported that they couldn’t reach **www.yummyrecipesforme.com**. After the page tried to load, they saw the error *“destination port unreachable”*. I reproduced the problem, captured the traffic with **tcpdump** while loading the site, and analysed the log to find which protocol and service were affected.

Evidence: tcpdump log

```
13:24:32.192571 IP 192.51.100.15.52444 > 203.0.113.2.domain: 35084+ A?
yummyrecipesforme.com. (24)
13:24:36.098564 IP 203.0.113.2 > 192.51.100.15: ICMP 203.0.113.2
udp port 53 unreachable length 254

13:26:32.192571 IP 192.51.100.15.52444 > 203.0.113.2.domain: 35084+ A?
yummyrecipesforme.com. (24)
13:27:15.934126 IP 203.0.113.2 > 192.51.100.15: ICMP 203.0.113.2
udp port 53 unreachable length 320

13:28:32.192571 IP 192.51.100.15.52444 > 203.0.113.2.domain: 35084+ A?
yummyrecipesforme.com. (24)
13:28:50.022967 IP 203.0.113.2 > 192.51.100.15: ICMP 203.0.113.2
udp port 53 unreachable length 150
```

tcpdump capture of the DNS queries and ICMP error responses

Time	Source → Destination	Protocol	What happened
13:24:32.19	192.51.100.15:52444 → 203.0.113.2:53	DNS over UDP	A-record query (ID 35084) for yummyrecipesforme.co

Time	Source → Destination	Protocol	What happened
			m
13:24:36.10	203.0.113.2 → 192.51.100.15	ICMP	**udp port 53 unreachable**
13:26:32.19	192.51.100.15 → 203.0.113.2:53	DNS over UDP	Query retried
13:27:15.93	203.0.113.2 → 192.51.100.15	ICMP	**udp port 53 unreachable**
13:28:32.19	192.51.100.15 → 203.0.113.2:53	DNS over UDP	Query retried
13:28:50.02	203.0.113.2 → 192.51.100.15	ICMP	**udp port 53 unreachable**

Part 1: Summary of the problem

The UDP traffic shows that **the DNS server can't be reached on port 53**. Every DNS query sent over UDP got back an **ICMP "Destination Unreachable (port unreachable)"** error reading *"udp port 53 unreachable"*, not a DNS answer. This happened on all three attempts.

Port 53 is used by DNS, which translates domain names into IP addresses. Because the browser never received the website's IP address, it never got as far as sending the HTTPS request to the web server. So the website itself may be working fine; the failure happens earlier, at name resolution.

Most likely issue: the DNS service on 203.0.113.2 isn't running or isn't listening on UDP port 53, so the host rejects the packets.

Part 2: Analysis and likely cause

When it happened: The first failure was recorded at **13:24:32 (1:24 p.m.)** and the problem continued through at least 13:28:50.

How the IT team found out: Several of the client's customers reported that they couldn't reach the website, and the client passed this on to our IT services team.

How we investigated:

1. Visited the website to reproduce the error ("destination port unreachable").
2. Started tcpdump and loaded the page again to capture the traffic.
3. Analysed the capture: matched each outgoing DNS query to the ICMP response that came back, and noted the source and destination IPs, the ports and the timestamps.
4. Escalated the findings to my supervisor. Security engineers are now handling the incident.

Key findings:

- Client 192.51.100.15 sent DNS A-record queries for yummyrecipesforme.com to DNS server **203.0.113.2** on **UDP port 53**.
- The DNS server's host was up and reachable, since it sent back ICMP messages. However, it returned **ICMP port unreachable** every time, which means **no service was accepting traffic on port 53**.

- The error was consistent across three attempts over about four minutes, so this isn't a one-off packet loss.

Likely causes, most to least likely:

1. **The DNS service has stopped or crashed** on 203.0.113.2. For example, the DNS server software failed, or was stopped for maintenance and not restarted. This best fits a host that is up but has a closed port.
2. **A firewall or ACL is rejecting UDP/53**, perhaps because of a misconfigured rule or a recent change.
3. **A DoS attack against the DNS server** that has exhausted the service. This is less likely here, because the capture shows no unusual traffic volume.

Recommended next steps

- Check the DNS service status on 203.0.113.2 and restart it if needed. Review its logs for the cause of the failure.
- Review recent firewall and ACL changes that affect UDP (and TCP) port 53.
- Confirm that a secondary DNS server is configured and working, so one failure can't take the website offline again.
- Add monitoring and alerts for DNS service availability, so the team finds out before customers do.

Reflection

The key lesson was reading the log by layer. The ICMP reply proved that the DNS host was reachable at the network layer. The *port-unreachable* code pointed to a problem at the transport and application layer: the service, not the network.